# INNOVATION CHALLENGE PROJECT PLAN

The Altec Innovation Challenge is a **Student Led Contest,** and as such the Project Plan and Presentations should be completed by the students. Teachers and Volunteers are to serve as resources and guides. This document is designed to be a framework to help organize and guide the planning of the project. It will also serve as a scoring rubric to determine which teams move to the finals. The deadline for turning in this document is **October 3, 2025.**

**School Name: Burton Center for Arts & Technology**

**Team Name: Mystical Six**

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**Volunteer (if applicable):** N/A

| Student Name | Role | Grade |
|---|---|---|
| Cameron Chidester | Team Lead | 11 |
| Aiden Nowocin | Budgeting & Research | 11 |
| Joshua Ebacha | Design Team Lead (Design Team) | 11 |
| Jeremiah Bellamy | Hardware Developer (Design Team) | 11 |
| Wesley Owen | Coding Developer (Design Team) | 11 |
| Max Lang | Outreach Representative | 11 |

# Project Summary (10 points)

**Problem Description (250-word limit)**

Many teens at our schools reported being tired and drowsy on their way back from practices and school events. Many teens and adults also report being tired and drowsy on their drives to work and school. Unfortunately, even with the recommended 8 hours of sleep at night, a lot of people still find themselves drowsy and sleepy while driving. Drowsy driving is a factor in a substantial number of crashes, with one study showing 17.6% of all fatal crashes from 2017–2021 involved a drowsy driver, a higher percentage than officially reported. Drowsy driving contributes to thousands of deaths in the U.S. each year, with estimates ranging from about 6,400 deaths annually according to the AAA Foundation for Traffic Safety. This is a massive problem in our society, and for everyday drivers without access to high level technology this problem must be addressed.

**Solution Summary (250-word limit)**

Creating a high vibrating wrist band which reads and tracks the user's heart rate and is programmed to set off a vibration over 50hz when the heart rate drops more than the driver's average resting rate. Before driving, the user will set their average resting rate by pressing one of the buttons. The band itself will have a 3D printed box that is meant to be placed on the lower wrist of either hand, within the box contains the motor that causes the vibrations. The band which holds the box in place is made of nylon. Vibrations in modern vehicles enhance safety by providing haptic feedback, or tactile alerts, that inform drivers of potential dangers without requiring visual or auditory distractions. These systems are an integral part of Advanced Driver Assistance Systems (ADAS) and can be felt through the steering wheel, driver's seat, and accelerator pedal. But unfortunately, these systems are not present in older cars which a large percentage of drivers, often new teens drivers have. The creation of this wristband will make these advanced technologies available and affordable to everyday drivers.

## Section I: (20 points)

**Ideation & Research**

**Explain how the team identified the problem and developed the solution.**

**A) Why is this issue important to you?**

Upon realizing all our group members who drove had the common issue of feeling signs of drowsiness after a long day of school, practices, or an event like a long day at the beach. We conducted further interviews with other members of our class; and found that an overwhelming number of students and adults have this same issue. We went further by conducting a small-scale voluntary survey and determined that about 89% of students felt signs of drowsiness while driving. This left us wondering what we could do to fix this dangerous situation. Although most websites claim the best way to fix drowsy driving is simply to sleep, this isn't quite the reality. The problem isn't being tired, the problem is the lack of brain function and zoning out that happens while driving, especially at night. Additionally, my team and I observed that sleeping before driving leads to one wanting more sleep thus falling asleep on the wheel. All these issues show how problematic this problem is, we need a way to have subtle motions that keep us awake. It is important to be 100 percent alert while driving, and unless you have a top-of-the-line new car, you unfortunately have no drowsy driving technology. This issue needs to be changed, and we must be alert and, in the moment, while driving.

**B) What brainstorming and research methods were used? Provide details on the research conducted related to the problem and the solution.**

Our team brainstormed methods to keep drowsy drivers awake based around the five senses. To make the collection of research easier, we all took on one of the five senses to research with the sixth person researching laws related to producing devices that keep people awake and devices that already exist that fit our solution. Once we all finished our research, we came back together with what we gathered. For sight we gathered that blue lights, cool lights, and any bright lights tend to keep people up. For hearing we gathered that noises from 30 decibels to 70 or higher decibels keep people up as well as melodic sounds or sounds that you would commonly associate with waking up like the sound your alarm makes in the morning. For touch we gathered that vibrations and changes in temperature are factors that keep people awake. For smell we gathered that the smell of peppermint, citrus, and eucalyptus all can keep people awake. For taste we gathered that spicy, sour, or really any extreme flavors tend to keep people awake. The research that we gathered on laws did not return any significant laws on devices, and there were minimal products that keep you awake and match our solution, though drivers like truck drivers have devices that if you're drowsy.

**C) What sources and resources were consulted? Describe how this research influenced your solution.**

After researching the different senses and their effect on drowsiness, either causing sleepiness or waking the driver up was considered. Physical sense such as vibrations and auditory senses alert drivers, this can be seen in rumble strips on the side of highways. When a car begins to float to the end of the road, long protruded strips of concrete cause the car to vibrate, alerting the driver they are about to run off the run. Following the concept of vibrations alerting drivers, we designed and programmed a wrist band which either constantly vibrates every few minutes or vibrates when your heart rate drops too much. Reading the heart rate is important, when falling asleep your heart rate drops around 20 to 30 percent of your resting heart rate. Tracking heart rate is a good way to indicate drowsiness, and a small vibration alerts the driver.

## Section II: (30 points)

**SWOT Analysis of Identified Solution**

List three Strengths of the Solution:

1. The device monitors heart rate to detect a sign of drowsiness.
2. The solution uses vibration to alert the driver.
3. The user controls the device by setting the baseline and timer.

List three Weaknesses of the Solution:

1. A user-set heart rate baseline can be inaccurate due to many things.
2. Users may get used to the vibration reducing the effects.
3. The sensor is small and cheap so it may be inaccurate if we don't test it enough.

List three Opportunities the Solution would provide:

1. Safer roads, especially in the nighttime and morning hours.
2. Increased attentiveness and focus while driving.
3. Reduces road accidents and vehicle damage.

List three Potential Threats to the Solution:

1. Vibrations on one's wrist might cause minor unintended driving adjustments.
2. Vibrations might cause some people to experience nausea or motion sickness.
3. The motions or sound of a vibrating device might distract the user from the road.

Countermeasures: For each threat listed above, describe at least one action your team could take to reduce or overcome the risk.

The vibrations may cause the driver to unintentionally have to adjust, causing an unwanted movement of the hand. This is a big concern for our project and extensive research has gone into this issue. After researching the effect of the vibration, it will not cause the driver to have a large movement of their hand that will cause them to crash. If the band is placed on top of the wrist the driver isn't in danger, the hz level of the vibration also isn't enough to be harmful. It could even better help the driver stay alert, if the driver is constantly having to make small adjustments, then they must stay alert to do so.

## Section III: (30 points)

### Resources/Budget

*It's important to ensure you have the needed resources to accomplish the task. Create a list of all the resources you think you will need i.e. money, tools, knowledge/expertise, space, etc. Can you implement the solution with the $10,000? Will you need to seek additional funding? Research and shop around to get estimates of best pricing.*

**Budget: $10,000**

| | |
|---|---|
| **1. Arduino Pro Mini 5v/16MHz** | **$4.66** |
| **2. Pulse Sensor** | **$25.00** |
| **3. Micro Tiny 612 Vibration Motor** | **$1.67** |
| **4. 9V Battery Holder** | **$3.00** |
| **5. 1kΩ Resistor** | **$0.05** |
| **6. NPN Transistor** | **$0.30** |
| **7. Jumper Wires** | **$0.10** |
| **8. Diode** | **$0.05** |
| **9. Wrist Band** | **$1.92** |
| **10. Green Micro LED** | **$0.06** |
| **11. Total Price Per Unit** | **$36.80** |

**Our solution will be implemented into 100+ people's everyday life. The product will be on a person's wrist mainly for driving but can be used anywhere such as work, school, or even your home.**

- **Tools:** Equipment, hardware, and software required to complete the project.

**We will need a multitude of small hardware such as LEDs, heart rate detectors, microcontrollers, batteries, wires, and connectors. Software we will need to use is the coding software compatible with our microcontroller which is Arduino IDE.**

- **Knowledge/Expertise:** Skills or guidance you will need from teachers, mentors, or experts in the field.

**We will need guidance from those who are well versed in computers and putting together computer components as well as guidance in coding.**

- **Other:** Any additional resources not covered above (e.g., transportation, partnerships, or permissions).
  **Approval by a driving or trucking company to implement our design into the community.**

## Section IV: (10 points)

**Timeline and Roles**

*Using the Timeline and Roles section describe HOW you plan to implement your solution during the Spring Semester. Create a timeline of steps and when you'll accomplish each specific action needed to implement the solution.*

*Who on your team is responsible for each task (Champion)?*

**Implementation Timeline:** *(be as detailed as possible and determine who will be responsible for the task)*

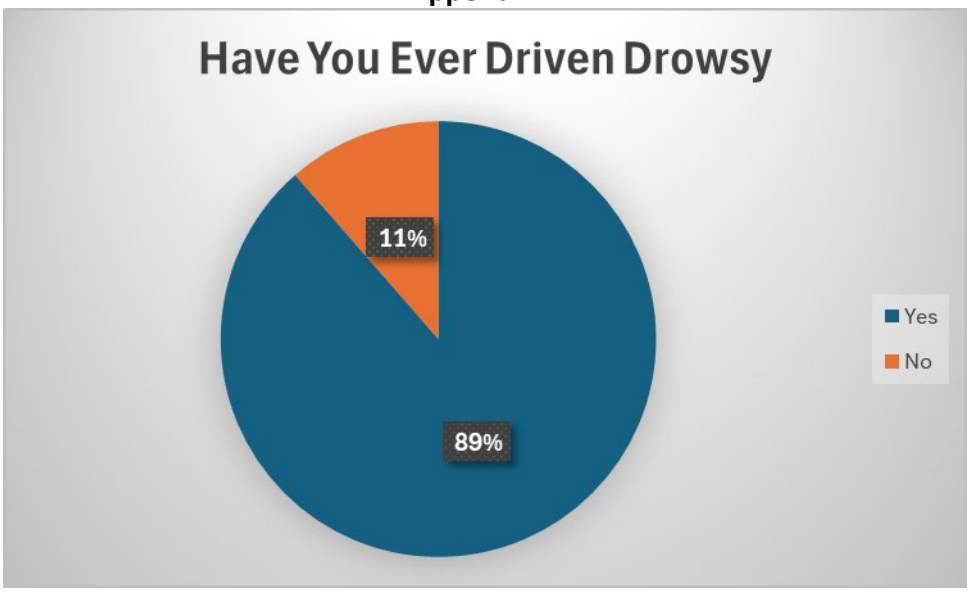
| ***Date*** | ***Task*** | ***Champion*** |
|---|---|---|
| *12-13-2025* | *Order parts for official prototype* | *Aiden Nowocin* |
| *12-22-2025* | *Assemble and test official prototype* | *Design Team* |
| *1-5-2026* | *Revise design if necessary and order new parts* | *Whole Team* |
| *1-13-2026* | *Assemble and test new prototype if needed* | *Design Team* |
| *1-21-2026* | *Make final revisions and order parts for all models* | *Design Team* |
| *2-20-2026* | *Assemble all models and prepare for exportation* | *Whole Team* |
| *3-1-2026* | *Finalize website to assist users* | *Wesley Owen* |
| *3-15-2026* | *Finalize dates of implementation* | *Max Lang* |
| *4-2-2026* | *Give our product to those in need* | *Whole Team* |
| *4-14-2026* | *Conduct user surveys and plan design improvements* | *Cameron Chidester* |

**Appendix A**

Resources sheet

| URL | What We Learned |
|---|---|
| https://my.clevelandclinic.org/health/diagnostics/23429-heart-rate-monitor | How heart rate monitors work |
| https://www.amazon.com/HiLetgo-Atmega328P-AU-Development-Microcontroller-Bootloadered/dp/B00E87VWQW/ | Arduino micro processor |
| https://www.amazon.com/Battery-Holder-Switch-Cover-Storage/dp/B0B3Y8ZRYP/ | 9V Battery Holder |
| https://www.nature.com/articles/s41598-025-08582-2 | Article reporting how effective HRV devices are |
| https://www.amazon.com/PulseSensor-com-Original-Pulse-Sensor-project/dp/B01CPP4QM0/ | Pulse Sensor |
| https://www.amazon.com/Vibration-Waterproof-8000-20000RPM-Electronic-equipments/dp/B0CY2KLH2P | Vibration Motor |
| https://www.amazon.com/SparkFun-FTDI-Basic-Breakout-5V/dp/B0068QKQEA/ | FTDI Basic Breakout |
| https://www.amazon.com/California-JOS-Breadboard-Solderless-Distribution/dp/B0BRQX6G3T/ | Bread board |
| https://www.straps.co.nz/blogs/blog/durability-matters-the-best-watch-straps-for-everyday-wear | Material used for watch straps |
| https://www.watchstrapstyle.co.uk/blogs/news/the-best-watch-straps-for-sensitive-skin | Material used for watch straps that won't cause allergic reactions |
| https://heritela.com/best-hypoallergenic-watch-straps-for-sensitive-skin/ | Best materials for hypoallergenic watch straps |
| https://www.nhtsa.gov/risky-driving/drowsy-driving | NHTSA Research |
| https://www.amazon.com/Resistor-Tolerance-Resistors-Multiple-Resistance/dp/B0B4G8HQD1/ | 1kΩ Resistor |
| https://www.amazon.com/microtivity-Bipolar-Amplifier-Switching-Transistors/dp/B0092CRKDC/ | NPN Transistor |
| https://www.amazon.com/California-JOS-Breadboard-Optional-Multicolored/dp/B0BRTJQZRD/ | Jumper Wires |
| https://www.amazon.com/BOJACK-Rectifier-IN4001-Electronic-Silicon/dp/B07Q3HBM63/ | Diode |
| https://www.amazon.com/OWOFYDR-diffused-Electronic-Component-Indicator/dp/B09B9FD3Z3/ | Green Micro LED |

**Appendix B**




In a survey that would ask multiple people if they have ever driven while drowsy, 35 of the people asked 31 said yes and 4 said no. Majority of the demographic was teenagers who recently got their license, 3 people were interviewed.

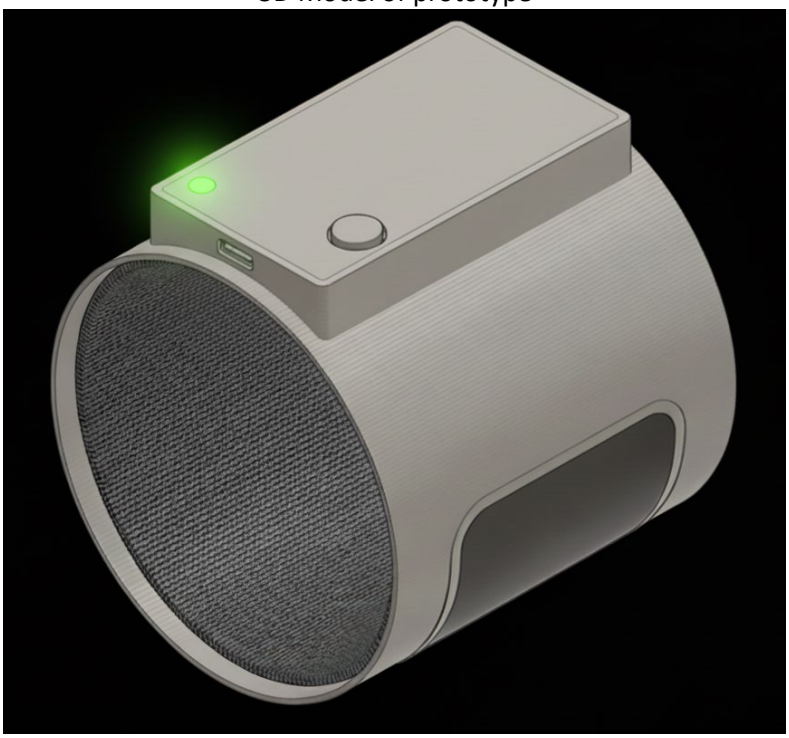
Stephen Neale: “Yesterday I worked at 5am, while driving I fell asleep at the stop light missing 3 cycles”.

Claire Yanishack: “One day I was working an eight-hour shift, when driving home I was nodding off. To stay awake I needed to blast music”.

Gage Allran: “While driving on the highway, I closed my eyes for a little too long and almost ran off the road”.

## Appendix C

First, install the Arduino IDE software on your computer and add the PulseSensor Playground library from the library manager. After wiring the pulse sensor and motor driver to the specified pins on your Arduino, connect the board to your computer and press the upload button in the IDE to transfer the code.

**Appendix D**

3D Model of prototype